

21. Scalability & Future Plan

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Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority
1	JSON flat-file storage is not safe under concurrent writes	Risk of data loss/corruption under multi-user load	High
2	No user authentication or authorization	Cannot track individual learning history or restrict access	High
3	Single-process Uvicorn server, no caching layer	Limited throughput under heavy concurrent traffic	Medium
4	Duplicate router-based and direct API endpoints	Increases maintenance overhead and API surface confusion	Medium

Scalability Plan

Scalability Aspect	Current State	Proposed Upgrade / Solution
User Load	Single-process Uvicorn, JSON files	Deploy behind Gunicorn + multiple Uvicorn workers; add a reverse proxy / load balancer
Data Storage	JSON flat files (data/*.json)	Migrate to PostgreSQL or MongoDB while preserving the existing db_manager.py interface
Performance	No caching; every request hits Gemini/T5 directly	Add response caching for repeated topics; batch quiz generation requests
Security	No auth, open CORS	Add JWT-based authentication, restrict CORS origins, encrypt sensitive data at rest

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Migrate JSON storage to a relational database (PostgreSQL)	Next 1–2 months	Safe concurrent access, richer querying/reporting
Phase 3	Add user authentication and per-user learning history / progress tracking	Next 2–3 months	Personalized dashboards, quiz score history
Phase 4	Consolidate duplicate API endpoints; add caching and rate limiting	Next 3–4 months	Cleaner API surface, better performance under load
Phase 5	Public deployment with CI/CD pipeline and analytics dashboard	Next 4–6 months	Production-ready, measurable platform usage